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College of Science and Technology
Rinchending: Bhutan

WEB101
Web Application Fundamentals
(SS2026)

Practical 1 Report

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RUB Wheel of Academic Law: Academic Dishonesty

Section H2 of the Royal University of Bhutan's *Wheel of Academic Law* provides the following definition of academic dishonesty:

Academic dishonesty may be defined as any attempt by a student to gain an unfair advantage in any assessment. It may be demonstrated by one of the following:

1. **Collusion:** the representation of a piece of unauthorised group work as the work of a single candidate.
2. **Commissioning:** submitting an assignment done by another person as the student's own work.
3. **Duplication**: the inclusion in coursework of material identical or substantially similar to material which has already been submitted for any other assessment within the University.
4. **False declaration**: making a false declaration to receive special consideration by an Examination Board or to obtain extensions to deadlines or exemption from work.
5. **Falsification of data**: presentation of data in laboratory reports, projects, etc., based on work purported to have been carried out by the student, which has been invented, altered or copied by the student.
6. **Plagiarism**: the unacknowledged use of another's work as if it were one's own.

Examples are:

- verbatim copying of another's work without acknowledgement.
- paraphrasing of another's work by simply changing a few words or altering the order of presentation, without acknowledgement.
- ideas or intellectual data in any form presented as one's own without acknowledging the source(s).
- making significant use of unattributed digital images such as graphs, tables, photographs, etc., taken from test books, articles, films, plays, handouts, the internet, or any other source, whether published or unpublished.
- submission of a piece of work which has previously been assessed for a different award or module or at a different institution as if it were new work.
- use of any material without prior permission of copyright from the appropriate authority or owner of the materials used".

Aim

To build a complete stack web-application which replicates the overall structure of the UI and the navigation path of TikTok by making use of Next.js, React and Tailwind CSS. In addition to developing an application that demonstrates how to set up a new application, develop your component's architecture, develop the layout for your app, implement the video feed, implement the user authentication form(s), and validate all of the input from your user on the client-side.

Theory

Next.js

Next.js is an open-source React framework developed by Vercel that enables developers to build production-grade web applications with features such as server side rendering (SSR), static site generation (SSG), and the App Router. Unlike a plain React application, Next.js provides a structured file-based routing system where each file inside the app/ directory automatically becomes a route, eliminating the need for third-party routing libraries.

React & JSX

React is a library that helps us build the parts of a website that people see. We use something called components to make it easier to work with the code. These components are like boxes that can have their own information and they can also get information from other parts of the website. We will be using something called JSX with our JavaScript files. JSX is nice because it looks a lot, like the HTML code that we use to make websites.

Tailwind CSS

Tailwind CSS is a utility-first CSS framework that enables developers to style elements directly in JSX using predefined class names such as flex, grid, p-4, text-xl, and bg-red-500. Rather than writing custom CSS, these utilities are composed directly in the markup, enabling rapid and consistent UI development.

React Hook Form

React Hook Form is a performant library for managing form state and validation in React. It minimizes unnecessary re-renders by using uncontrolled components and a ref-based approach.

Project Folder Structure

The project follows the Next.js App Router convention with a structured src/ directory:

```
src/
    app/                    ← Route pages (profile, upload, login,signup, etc..)
    components/
        layout/             ← Shared layout (MiainLayoutjsx)
        ui/                 ← Reusable UI components
    lib/                    ← Utility functions and helpers
```

Implementation Steps

Part 1 - Getting Started

1. Created a new Next.js project using command:
   npx create-next-app@latest

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\DELL\OneDrive\Desktop\WEB101_02240368_TikTok>
PS C:\Users\DELL\OneDrive\Desktop\WEB101_02240368_TikTok> npx create-next-app@latest
Need to install the following packages:
create-next-app@16.2.0
Ok to proceed? (y) y
√ What is your project named? ... my-app
√ Would you like to use the recommended Next.js defaults? » No, customize settings
√ Would you like to use TypeScript? ... No / Yes
√ Which linter would you like to use? » ESLint
√ Would you like to use React Compiler? ... No / Yes
√ Would you like to use Tailwind CSS? ... No / Yes
√ Would you like your code inside a `src/` directory? ... No / Yes
√ Would you like to use App Router? (recommended) ... No / Yes
√ Would you like to customize the import alias (`@/*` by default)? ... No / Yes
```

2. Cleaned up the default project
   Navigate to the src/app directory
   Replace the contents of page.js with a basic component:

```
export default function Home() {
  return (
    <main className="flex min-h-screen flex-col items-center justify-center p-24">
      <h1 className="text-4xl font-bold">TikTok</h1>
      <p className="mt-4 text-xl">Welcome to our TikTok project!</p>
    </main>
  );
}
```

   Clean up globals.css in the app directory to keep only the Tailwind directives:
   @import "tailwindcss";

```
page.jsx    README.md    Reflection.md    # globals.css ×
my-app > src > app > # globals.css
1    @import "tailwindcss";
```

3. Set Up Project Structure
   Create these folders to organize your project (inside the src directory):

```
PS C:\Users\DELL\OneDrive\Desktop\WEB101_02240368_TikTok\my-app> mkdir -p src/components/layout

    Directory: C:\Users\DELL\OneDrive\Desktop\WEB101_02240368_TikTok\my-app\src\components

Mode                LastWriteTime         Length Name
----                -------------         ------ ----
d-----        3/19/2026  10:53 PM                layout

PS C:\Users\DELL\OneDrive\Desktop\WEB101_02240368_TikTok\my-app> mkdir -p src/components/ui

    Directory: C:\Users\DELL\OneDrive\Desktop\WEB101_02240368_TikTok\my-app\src\components

Mode                LastWriteTime         Length Name
----                -------------         ------ ----
d-----        3/19/2026  10:53 PM                ui

PS C:\Users\DELL\OneDrive\Desktop\WEB101_02240368_TikTok\my-app> mkdir -p src/lib

    Directory: C:\Users\DELL\OneDrive\Desktop\WEB101_02240368_TikTok\my-app\src

Mode                LastWriteTime         Length Name
----                -------------         ------ ----
d-----        3/19/2026  10:53 PM                lib

PS C:\Users\DELL\OneDrive\Desktop\WEB101_02240368_TikTok\my-app> mkdir -p src/app/upload

    Directory: C:\Users\DELL\OneDrive\Desktop\WEB101_02240368_TikTok\my-app\src\app
```

4. Create Basic Layout Components

Create a basic layout component in src/components/layout/MainLayout.jsx with a sticky header containing navigation links to Home, Profile, and Upload pages, wrapping all page content in a flexible main container.

```jsx
import Link from 'next/link';

export default function MainLayout({ children }) {
  return (
    <div className="flex flex-col min-h-screen">
      <header className="sticky top-0 z-50 bg-white border-b">
        <div className="flex items-center justify-between h-16 px-4">
          <Link href="/" className="text-xl font-bold">
            TikTok
          </Link>
          <nav>
            <ul className="flex space-x-4">
              <li>
                <Link href="/" className="hover:text-red-500">
                  Home
                </Link>
              </li>
              <li>
                <Link href="/profile" className="hover:text-red-500">
                  Profile
                </Link>
              </li>
              <li>
                <Link href="/upload" className="hover:text-red-500">
                  Upload
                </Link>
              </li>
            </ul>
          </nav>
        </div>
      </header>
      <main className="flex-grow">
        {children}
      </main>
    </div>
  );
}
```

5. Update layout.js in the App Directory

Modify src/app/layout.js to include your layout:

```jsx
import { Inter } from 'next/font/google';
import './globals.css';
import MainLayout from '@/components/layout/MainLayout';

const inter = Inter({ subsets: ['latin'] });

export const metadata = {
  title: 'TikTok Clone',
  description: 'A TikTok clone built with Next.js',
};

export default function RootLayout({ children }) {
  return (
    <html lang="en">
      <body className={inter.className}>
        <MainLayout>
          {children}
        </MainLayout>
      </body>
    </html>
  );
}
```

6. Create Basic Pages for Navigation

Create a profile page at src/app/profile/page.jsx:

```
export default function ProfilePage() {
  return (
    <div className="container mx-auto p-4">
      <h1 className="text-2xl font-bold mb-4">User Profile</h1>
      <div className="bg-gray-100 p-6 rounded-lg">
        <div className="w-20 h-20 bg-gray-300 rounded-full mb-4"></div>
        <h2 className="text-xl font-semibold">Username</h2>
        <p className="text-gray-600">@username</p>
        <div className="mt-4 flex space-x-4">
          <div>
            <span className="font-bold">0</span> Following
          </div>
          <div>
            <span className="font-bold">0</span> Followers
          </div>
          <div>
            <span className="font-bold">0</span> Likes
          </div>
        </div>
      </div>
    </div>
  );
}
```

7. Create Basic Pages for Navigation

Create a profile page at src/app/profile/page.jsx:

```
export default function ProfilePage() {
  return (
    <div className="container mx-auto p-4">
      <h1 className="text-2xl font-bold mb-4">User Profile</h1>
      <div className="bg-gray-100 p-6 rounded-lg">
        <div className="w-20 h-20 bg-gray-300 rounded-full mb-4"></div>
        <h2 className="text-xl font-semibold">Username</h2>
        <p className="text-gray-600">@username</p>
        <div className="mt-4 flex space-x-4">
          <div>
            <span className="font-bold">0</span> Following
          </div>
          <div>
            <span className="font-bold">0</span> Followers
          </div>
          <div>
            <span className="font-bold">0</span> Likes
          </div>
        </div>
      </div>
    </div>
  );
}
```

8. Create an upload page at src/app/upload/page.jsx:

```
export default function UploadPage() {
  return (
    <div className="container mx-auto p-4">
      <h1 className="text-2xl font-bold mb-4">Upload Video</h1>
      <div className="bg-gray-100 p-6 rounded-lg">
        <div className="border-2 border-dashed border-gray-300 p-12 text-center rounded-lg">
          <p className="text-gray-500">
            Drag and drop a video file
          </p>
          <button className="mt-4 bg-red-500 text-white px-4 py-2 rounded">
            Select File
          </button>
        </div>
      </div>
    </div>
  );
}
```

9. Start the Development Server

Your project should now be running at http://localhost:3000

Part 2 - Creating the Web Layout and Main Interface

1. Installed the react-icons package.

   npm install react-icons

```
PS C:\Users\DELL\OneDrive\Desktop\WEB101_02240368_TikTok> cd my-app
PS C:\Users\DELL\OneDrive\Desktop\WEB101_02240368_TikTok\my-app> npm install react-icons

added 1 package, and audited 357 packages in 20s

143 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
PS C:\Users\DELL\OneDrive\Desktop\WEB101_02240368_TikTok\my-app>
```

2. Update the Layout Structure

   Let's restructure our layout to match TikTok's web version with a sidebar. Update src/components/layout/MainLayout.jsx:

```
'use client';
import Link from 'next/link';
import {
  FaHome, FaUserFriends, FaCompass, FaVideo,
  FaInbox, FaRegUser, FaPlus
} from 'react-icons/fa';

export default function MainLayout({ children }) {
  return (
    <div className="flex min-h-screen">
      {/* Sidebar */}
      <div className="w-60 border-r fixed h-full overflow-y-auto">
        <div className="p-4">
          <Link href="/" className="text-xl font-bold flex items-center">
            <span className="text-red-500 mr-1">TikTok</span>
          </Link>
        </div>

        <nav className="mt-4">
          <ul className="space-y-2">
            <li>
              <Link
                href="/"
                className="flex items-center p-3 hover:bg-gray-100 rounded-md mx-2"
              >
                <FaHome className="text-xl mr-3" />
                <span>For You</span>
              </Link>
            </li>
            <li>
              <Link
                href="/following"
                className="flex items-center p-3 hover:bg-gray-100 rounded-md mx-2"
              >
                <FaUserFriends className="text-xl mr-3" />
                <span>Following</span>
              </Link>
```

```
            </li>
            <li>
              <Link
                href="/explore"
                className="flex items-center p-3 hover:bg-gray-100 rounded-md mx-2"
              >
                <FaCompass className="text-xl mr-3" />
                <span>Explore</span>
              </Link>
            </li>
            <li>
              <Link
                href="/live"
                className="flex items-center p-3 hover:bg-gray-100 rounded-md mx-2"
              >
                <FaVideo className="text-xl mr-3" />
                <span>LIVE</span>
              </Link>
            </li>
          </ul>
        </nav>

        <div className="border-t mt-4 pt-4 px-2">
          <p className="text-gray-500 text-sm px-3 mb-2">Suggested accounts</p>
          {Array.from({ length: 5 }).map(( _, index) => (
            <div key={index} className="flex items-center p-2 hover:bg-gray-100 rounded-md">
              <div className="h-8 w-8 rounded-full bg-gray-300 mr-2"></div>
              <div>
                <p className="text-sm font-semibold">user_{index + 1}</p>
                <p className="text-xs text-gray-500">User {index + 1}</p>
              </div>
            </div>
          ))}
        </div>
```

```
              <div className="h-8 w-8 rounded-full bg-gray-300 mr-2"></div>
              <div>
                <p className="text-sm font-semibold">user_{index + 1}</p>
                <p className="text-xs text-gray-500">User {index + 1}</p>
              </div>
            </div>
          ))}
        </div>

        <div className="px-3 py-4 mt-2">
          <p className="text-sm text-gray-500 mb-4">
            Log in to follow creators, like videos, and view comments.
          </p>
          <button className="w-full py-2 px-4 border rounded-md font-medium mb-2">
            Login
          </button>
        </div>

        <div className="border-t px-3 py-4 text-xs text-gray-500">
          <p class="mb-2">© 2025 TikTok</p>
        </div>
      </div>
```

```

      {/* Main content */}
      <div className="ml-60 flex-1">
        <div className="max-w-[1150px] mx-auto">
          {/* Top header with search */}
          <header className="h-16 border-b flex items-center justify-between px-4">
            <div className="w-1/3"></div>
            <div className="w-1/3">
              <div class="relative">
                <input
                  type="text"
                  placeholder="Search accounts and videos"
                  className="w-full bg-gray-100 py-2 pl-10 pr-4 rounded-full"
                />
                <FaCompass className="absolute left-4 top-1/2 transform -translate-y-1/2 text-gray-400" />
              </div>
            </div>
            <div class="w-1/3 flex justify-end space-x-4">
              <Link href="/upload">
                <button class="border px-3 py-1 rounded-md hover:bg-gray-50 flex items-center">
                  <FaPlus class="mr-2" /> Upload
                </button>
              </Link>

              <button class="bg-red-500 text-white px-6 py-1 rounded-md">
                Log in
              </button>
            </div>
            <div>
              <Link
                href="/profile"
                class="flex items-center p-3 hover:bg-gray-100 rounded-md mx-2"
              >
                <FaRegUser class="text-xl mr-3" />
                <span>Profile</span>
              </Link>
            </div>
          </header>
```

```

          {/* Main content */}
          <main>{children}</main>
        </div>
      </div>
    </div>
  );
}
```

3. Create a For You Page Component

Let's create a feed similar to TikTok's web version. Create a new file src/components/ui/VideoCard.jsx:

```jsx
'use client';
import { useState } from 'react';
import {
  FaHeart, FaRegHeart, FaComment,
  FaShare, FaMusic
} from 'react-icons/fa';

export default function VideoCard({ post }) {
  const [liked, setLiked] = useState(false);

  // Placeholder data
  const { username, caption, audio, likes, comments, shares } = post;

  const handleLikeClick = () => {
    setLiked(!liked);
  };

  return (
    <div class="flex py-6 border-b">
      {/* User avatar */}
      <div class="mr-3">
        <div class="h-12 w-12 rounded-full bg-gray-300"></div>
      </div>

      <div class="flex-1">
        {/* User info and caption */}
        <div class="mb-2">
          <span class="font-bold hover:underline cursor-pointer">{username}</span>
          <span class="text-sm ml-1">• 2d ago</span>
          <p class="text-sm mt-1">{caption}</p>
        </div>

        {/* Audio info */}
        <div class="flex items-center text-sm mb-3">
          <FaMusic class="mr-2 text-xs" />
          <span class="truncate max-w-[250px]">{audio}</span>
        </div>

        <div class="flex">
          {/* Video container */}
          <div class="mr-5 w-[300px] h-[530px] bg-black rounded-md flex items-center justify-center relative overflow-hidden">
            <p class="text-white">Video Placeholder</p>
            <div class="absolute bottom-4 left-4 text-white text-sm">
              <p class="mb-1">@:30</p>
            </div>
          </div>
```

```jsx
          {/* Interaction buttons */}
          <div class="flex flex-col justify-end space-y-3 py-2">
            {/* Like button */}
            <button
              class="flex flex-col items-center"
              onClick={handleLikeClick}
            >
              <div class="h-10 w-10 rounded-full bg-gray-200 flex items-center justify-center">
                {liked ? (
                  <FaHeart class="text-red-500" />
                ) : (
                  <FaRegHeart />
                )}
              </div>
              <span class="text-xs mt-1">{liked ? likes + 1 : likes}</span>
            </button>

            {/* Comment button */}
            <button class="flex flex-col items-center">
              <div class="h-10 w-10 rounded-full bg-gray-200 flex items-center justify-center">
                <FaComment />
              </div>
              <span class="text-xs mt-1">{comments}</span>
            </button>

            {/* Share button */}
            <button class="flex flex-col items-center">
              <div class="h-10 w-10 rounded-full bg-gray-200 flex items-center justify-center">
                <FaShare />
              </div>
              <span class="text-xs mt-1">{shares}</span>
            </button>
          </div>
        </div>
      </div>
    </div>
  );
}
```

4. Create a VideoFeed Component
   Create a new file src/components/ui/VideoFeed.jsx:

```jsx
'use client';
import VideoCard from './VideoCard';

// Sample data for our feed
const DUMMY_POSTS = [
  {
    id: '1',
    username: '@user1',
    caption: 'Check out this cool video! #trending #tiktok #viral',
    audio: 'Original Sound - User1',
    likes: 1234,
    comments: 432,
    shares: 89
  },
  {
    id: '2',
    username: '@user2',
    caption: 'Learning to dance 💃 #dance #fun #trending',
    audio: 'Popular Song - Artist',
    likes: 5678,
    comments: 321,
    shares: 52
  },
  {
    id: '3',
    username: '@user3',
    caption: 'Beautiful sunset today! #nature #sunset #vibes',
    audio: 'Sunset Vibes - Chill Music',
    likes: 2468,
    comments: 135,
    shares: 46
  }
];


export default function VideoFeed() {
  return (
    <div className="max-w-[550px] mx-auto">
      {DUMMY_POSTS.map((post) => (
        <VideoCard key={post.id} post={post} />
      ))}
    </div>
  );
}
```

5. Update the Home Page Now, let's update our home page to display the video feed.
   Update src/app/page.js:

```jsx
import VideoFeed from '../components/ui/VideoFeed';

export default function Home() {
  return (
    <div className="py-4">
      <VideoFeed />
    </div>
  );
}
```

6. Create a Following Page Create a new file src/app/following/page.jsx:

```jsx
export default function FollowingPage() {
  return (
    <div className="max-w-[550px] mx-auto py-10 text-center">
      <h2 className="text-2xl font-bold mb-3">Follow accounts</h2>
      <p className="text-gray-500 mb-8">Follow accounts to see their latest videos</p>

      <div className="grid grid-cols-5 gap-4 mb-6">
        {Array.from({ length: 5 }).map((_, index) => (
          <div key={index} className="text-center">
            <div className="h-16 w-16 rounded-full bg-gray-300 mx-auto mb-2"></div>
            <p className="text-sm font-semibold">user_{index + 1}</p>
            <button className="mt-2 text-xs bg-red-500 text-white py-1 px-4 rounded-full">
              Follow
            </button>
          </div>
        ))}
      </div>

      <button className="border border-gray-300 text-gray-700 py-2 px-8 rounded-md font-medium">
        See more
      </button>
    </div>
  );
}
```

7. Create an Explore Page Create a new file src/app/explore/page.jsx

```jsx
export default function ExplorePage() {
  return (
    <div className="p-4">
      <h2 className="text-2xl font-bold mb-4">Explore</h2>

      <div className="mb-8">
        <h3 className="text-xl font-semibold mb-3">Trending Hashtags</h3>
        <div className="grid grid-cols-3 gap-4">
          {Array.from({ length: 6 }).map((_, index) => (
            <div
              key={index}
              className="aspect-video bg-gray-200 rounded-md flex flex-col items-center justify-center p-4"
            >
              <p className="font-bold text-lg">#Trending{index + 1}</p>
              <p className="text-sm text-gray-500">{(index + 1) * 1.5}M views</p>
            </div>
          ))}
        </div>
      </div>

      <div>
        <h3 className="text-xl font-semibold mb-3">Popular Videos</h3>
        <div className="grid grid-cols-4 gap-3">
          {Array.from({ length: 12 }).map((_, index) => (
            <div
              key={index}
              className="aspect-[9/16] bg-gray-300 rounded-md flex items-center justify-center"
            >
              <p className="text-sm">Video {index + 1}</p>
            </div>
          ))}
        </div>
      </div>
    </div>
  );
}
```

8. Create a Live Page
   Create a new file src/app/live/page.jsx:

```jsx
export default function LivePage() {
  return (
    <div className="p-4">
      <h2 className="text-2xl font-bold mb-4">LIVE</h2>

      <div className="grid grid-cols-3 gap-4">
        {Array.from({ length: 9 }).map((_, index) => (
          <div key={index} className="rounded-md overflow-hidden">
            <div className="aspect-video bg-gray-200 flex items-center justify-center">
              <p>LIVE Stream {index + 1}</p>
            </div>
            <div className="p-2 border border-t-0 rounded-b-md">
              <div className="flex items-center">
                <div className="h-8 w-8 rounded-full bg-gray-300 mr-2"></div>
                <div>
                  <p className="text-sm font-semibold">user_{index + 1}</p>
                  <p className="text-xs text-gray-500">Live stream title here</p>
                </div>
              </div>
              <div className="mt-2 text-xs text-red-500">
                {(index + 1) * 125} viewers
              </div>
            </div>
          </div>
        ))}
      </div>
    </div>
  );
}
```

9. Create an Upload Page
   Let's update our upload page to match TikTok's web interface.
   Update src/app/upload/page.jsx:

```jsx
export default function UploadPage() {
  return (
    <div class="max-w-4xl mx-auto p-8">
      <h1 class="text-2xl font-bold mb-6">Upload video</h1>

      <div class="flex">
        <div class="w-[360px] border-dashed border-2 border-gray-300 rounded-lg p-8 flex flex-col items-center justify-center text-center">
          <div class="w-16 h-16 rounded-full bg-gray-100 flex items-center justify-center mb-4">
            <span class="text-4xl text-gray-400">+</span>
          </div>
          <h3 class="text-lg font-semibold mb-2">Select video to upload</h3>
          <p class="text-sm text-gray-500 mb-4">Or drag and drop a file</p>
          <p class="text-xs text-gray-400 mb-4">MP4 or WebM</p>
          <p class="text-xs text-gray-400 mb-4">720x1280 resolution or higher</p>
          <p class="text-xs text-gray-400">Up to 10 minutes</p>
          <p class="text-xs text-gray-400">Less than 2 GB</p>
          <button class="mt-8 bg-red-500 text-white py-2 px-8 rounded-md">
            Select file
          </button>
        </div>

        <div class="flex-1 ml-8">
          <div class="mb-4">
            <label class="block text-sm font-medium mb-2">Caption</label>
            <input
              type="text"
              class="w-full p-2 border rounded-md"
              placeholder="Add a caption..."
            />
          </div>

          <div class="mb-4">
            <label class="block text-sm font-medium mb-2">Cover</label>
            <div class="h-20 bg-gray-200 rounded-md"></div>
          </div>

          <div class="mb-4">
            <label class="block text-sm font-medium mb-2">Who can view this video</label>
            <select class="w-full p-2 border rounded-md">
              <option>Public</option>
              <option>Friends</option>
              <option>Private</option>
            </select>
          </div>

          <div class="mb-4">
            <label class="block text-sm font-medium mb-2">Allow users to:</label>
            <div class="space-y-2">
              <div class="flex items-center">
                <input type="checkbox" id="comments" class="mr-2" />
                <label htmlFor="comments">Comment</label>
              </div>
              <div class="flex items-center">
                <input type="checkbox" id="duet" class="mr-2" />
                <label htmlFor="duet">Duet</label>
              </div>
              <div class="flex items-center">
                <input type="checkbox" id="stitch" class="mr-2" />
                <label htmlFor="stitch">Stitch</label>
              </div>
            </div>
          </div>

          <div class="flex space-x-3 mt-6">
            <button class="px-6 py-2 border rounded-md">Discard</button>
            <button class="px-6 py-2 bg-red-500 text-white rounded-md">Post</button>
          </div>
        </div>
      </div>
    )
}
```

10. Update the Profile Page

Let's improve our profile page to better match TikTok's web interface.

Update src/app/profile/page.jsx:

```jsx
import { FaEdit, FaUserPlus } from 'react-icons/fa';

export default function ProfilePage() {
  return (
    <div className="max-w-4xl mx-auto py-8">
      {/* Profile header */}
      <div className="flex items-start mb-8">
        <div className="h-24 w-24 rounded-full bg-gray-300 mr-4"></div>

        <div className="flex-1">
          <h1 className="text-xl font-bold mb-2">@username</h1>
          <h2 className="text-lg mb-4">User Name</h2>

          <button className="px-6 py-1.5 rounded-md border border-gray-300 font-medium flex items-center mr-3">
            <FaEdit className="mr-2" /> Edit profile
          </button>

          <div className="flex items-center mt-4 space-x-4">
            <p><span className="font-bold">0</span> Following</p>
            <p><span className="font-bold">0</span> Followers</p>
            <p><span className="font-bold">0</span> Likes</p>
          </div>

          <p className="mt-4 text-sm max-w-md">
            No bio yet.
          </p>
        </div>
      </div>

      {/* Video tabs */}
      <div class="border-b">
        <div class="flex">
          <button class="py-2 px-4 border-b-2 border-black font-medium">
            Videos
          </button>
          <button class="py-2 px-4 text-gray-500 font-medium">
            Liked
          </button>
        </div>
      </div>
```

```jsx
      {/* Videos grid */}
      <div class="py-4">
        <div class="grid grid-cols-6 gap-1">
          {/* Empty state for now */}
          <div class="col-span-6 py-20 text-center">
            <div class="max-w-md mx-auto">
              <h3 class="text-xl font-bold mb-2">Upload your first video</h3>
              <p class="text-gray-500 mb-8">Your videos will appear here</p>
              <button class="bg-red-500 text-white px-8 py-2 rounded-md font-medium">
                Upload now
              </button>
            </div>
          </div>
        </div>
      </div>
    </div>
  );
}
```

11. Test Your Application

Run your development server and test the application.

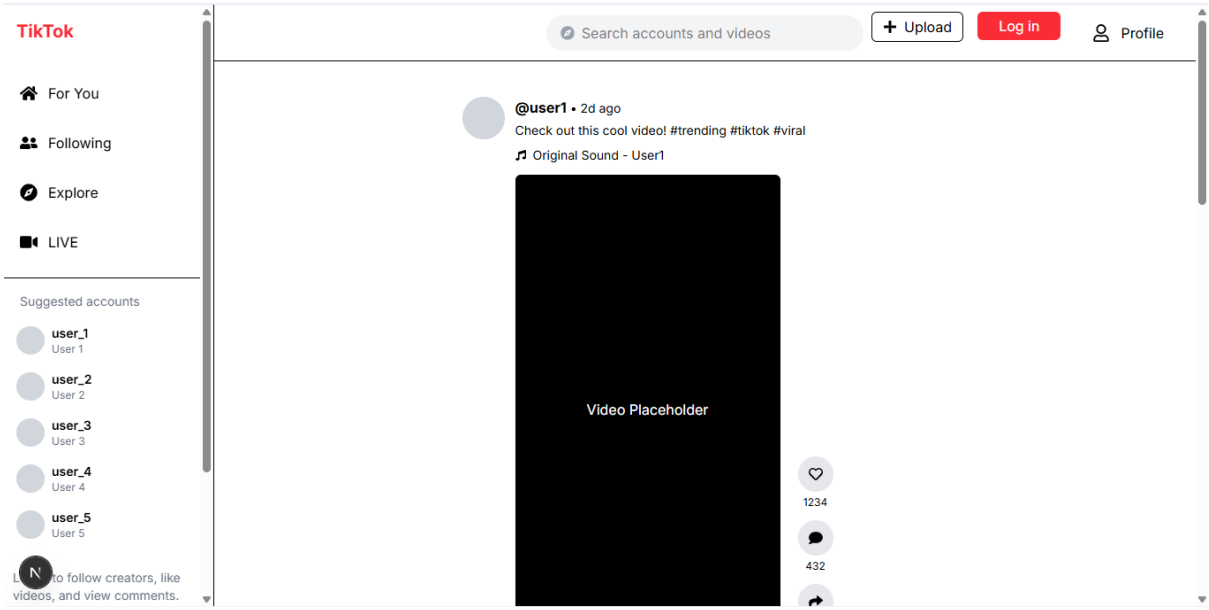
Part 3 - Creating Login and Registration Forms

Introduced form handling and validation using the React Hook Form library, creating fully validated Login and Signup pages.

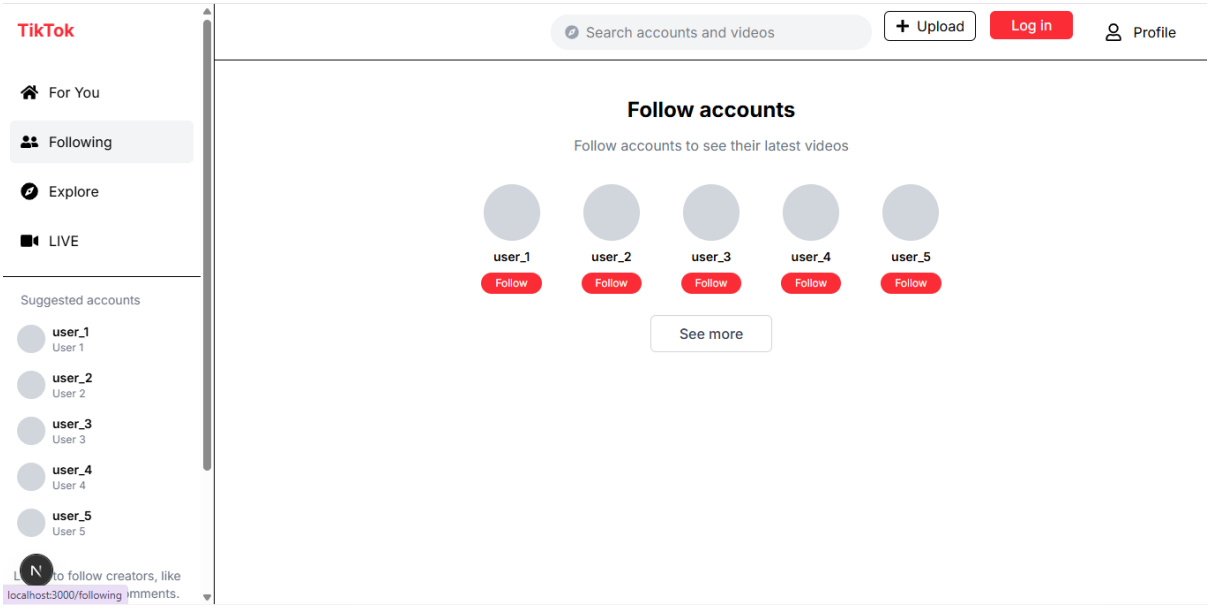
1. Installed React Hook Form using the command: npm install react-hook-form.
2. Created the Login page (src/app/login/page.jsx) as a client component. The useForm() hook was used to register email and password fields with required validation rules. On submission, handleSubmit triggers the onSubmit function which sets isLoading to true and simulates a 1500ms API call before displaying a success alert.
3. Created the Signup page (src/app/signup/page.jsx) with the following validated fields: username (required, minimum 3 characters, alphanumeric pattern only), email (required, validated against a regex pattern for proper email format), password (required, minimum 8 characters, must contain at least one uppercase letter, one lowercase letter, one number, and one special character), confirm password (required, uses a custom validate function that compares the value against the watched password field), and a terms and conditions checkbox (required).
4. Updated MainLayout.jsx with two sets of changes: first, the sidebar Login/Sign Up buttons were wrapped in Next.js Link components pointing to /login and /signup respectively; second, the top header Log in button was wrapped in a Link pointing to /login.
5. Updated MainLayout.jsx with two sets of changes: first, the sidebar Login/Sign Up buttons were wrapped in Next.js Link components pointing to /login and /signup respectively; second, the top header Log in button was wrapped in a Link pointing to /login.

OUTPUT:

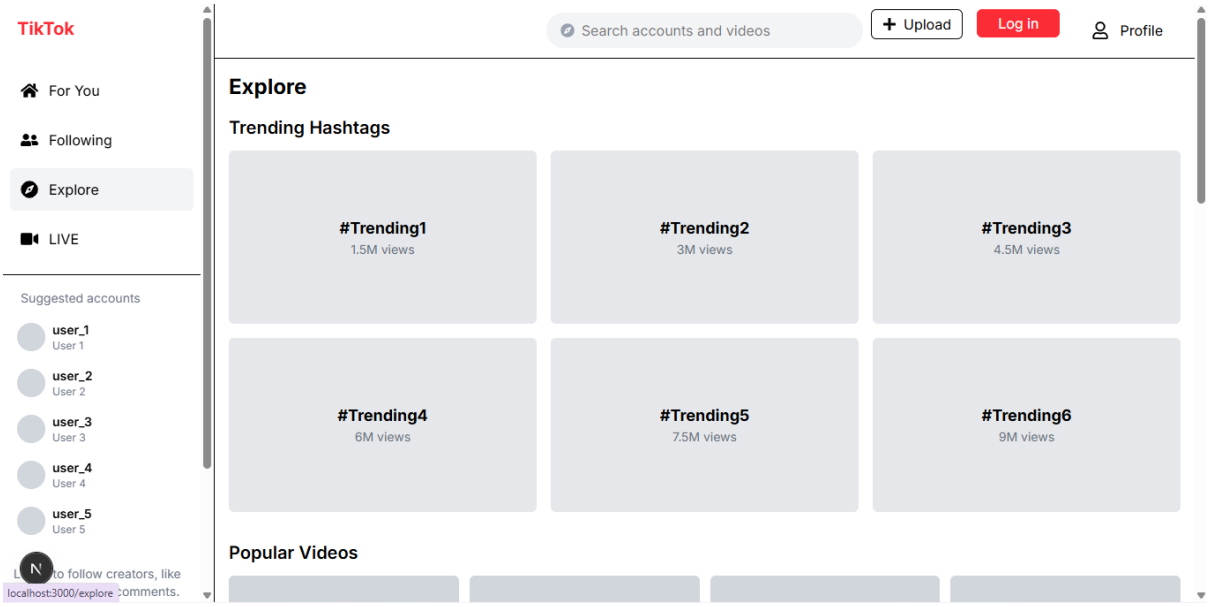
1.For You page



2. Following Page

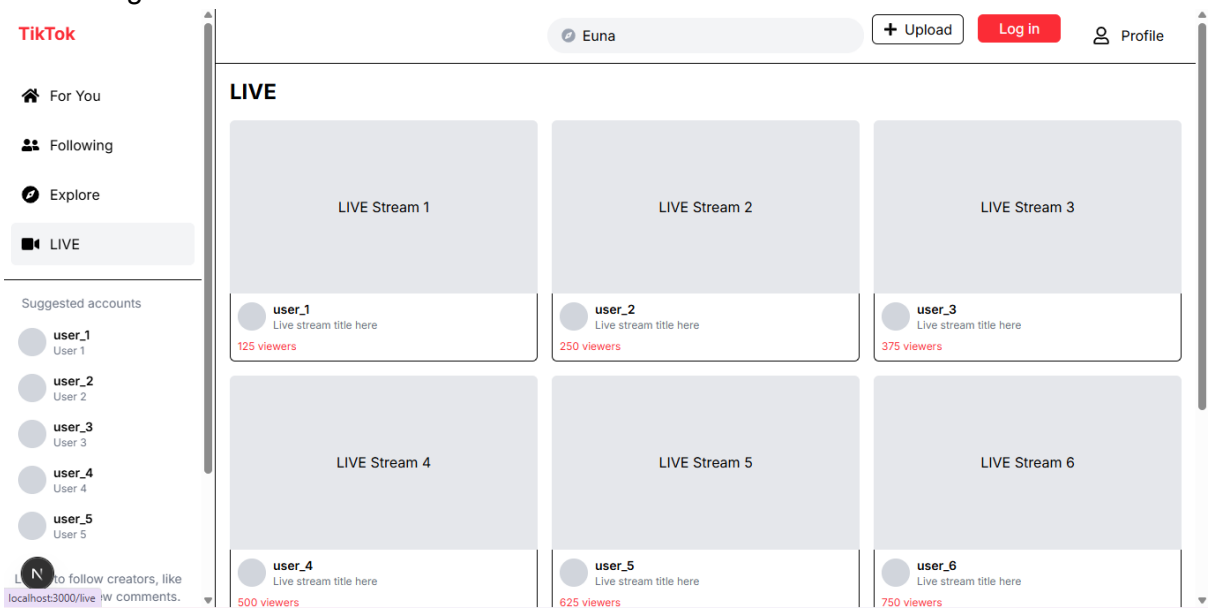


3. Explore Page

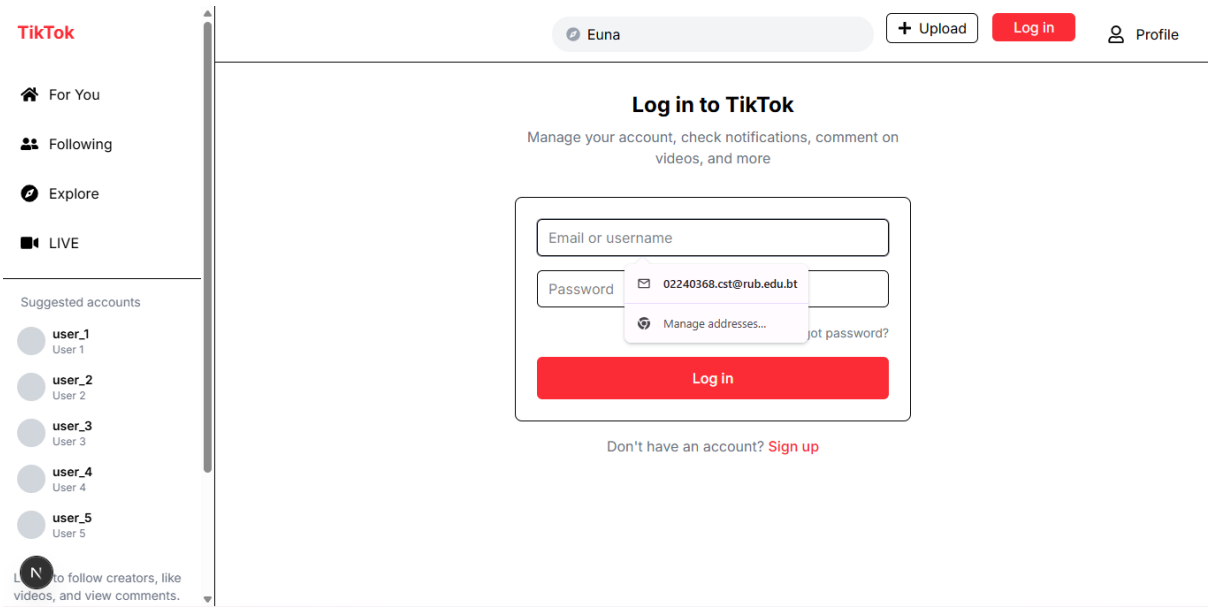


4. Search Bar

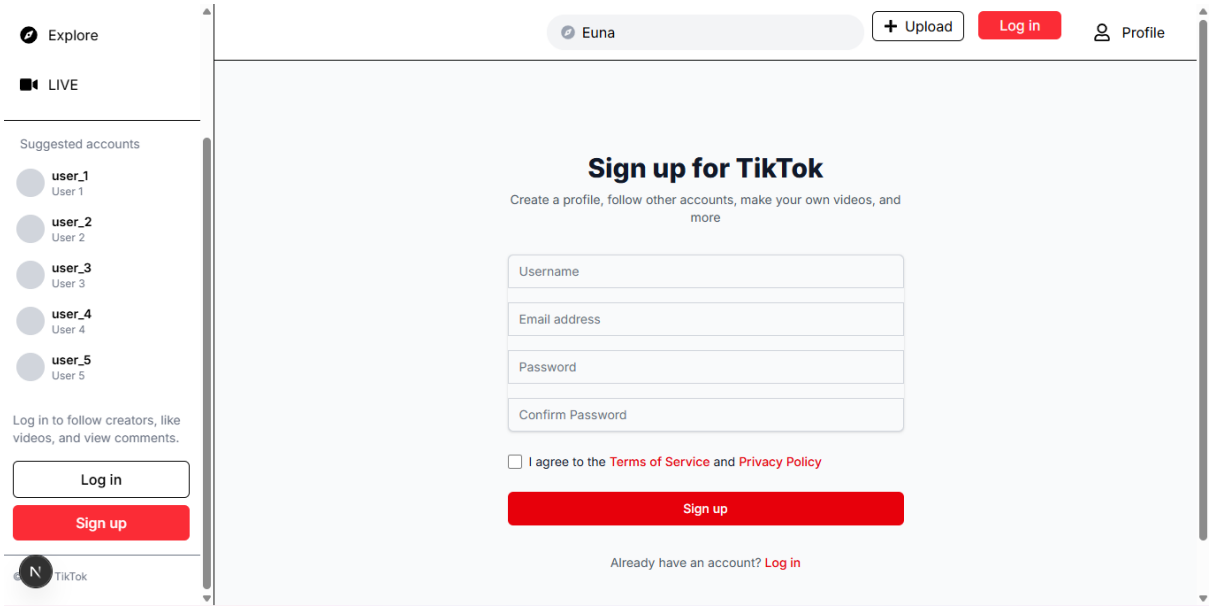
5.Live Page



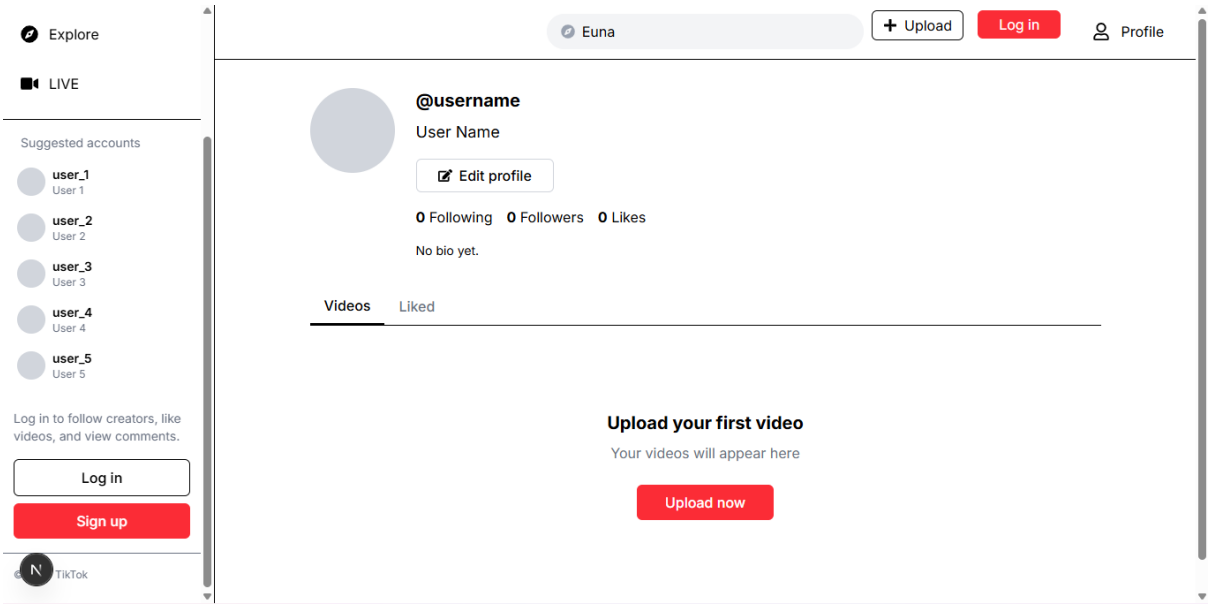
6. Login Page



7. SignUp Page



8. Profile Page



CONCLUSION:

This practical gave opportunity to build a modern multi-page web application from scratch. The whole application was built during the practical using Next.js, React and Tailwind CSS. In three parts the development of the application went step-by-step from initialising the project and planning the structure of the files and folders to the implementation of advanced UI components and client side validation of the forms.

Key learning outcomes achieved:
- Next.js App Router, the file-based routing system
- How layout.js provides a statically shared UI component that doesn't re-mount on every page change.

Practice Exercise: TikTok Website Decomposition Decompose the TikTok website into React components. Identify and list the class names for each component, and describe the props and state for each class. You may use any online resource to investigate the website structure and design. Be sure to explain your reasoning and provide key terms to identify each class. Your submission should not exceed the length specified. Focus your answers and avoid over-explaining, especially for brief inputs.

•Build responsive layouts with ease using Tailwind CSS utility classes, without having to write any custom CSS for layouts such as fixed sidebars, sticky headers, grid content areas, and form inputs.

•Using useState hook to manage the state of the application for example handling video like toggling and form loading states.

•Form Registration Using React Hook Form for form submission and form validation based on multiple rules like required, pattern, minLength using built-in methods and a custom validate function.

This practical encountered a few issues. At the start, it was difficult to work out the difference between Server Components and Client Components in the Next.js App Router. The Next.js documentation did not provide sufficient information. The fact that components which use useState or event listeners have to be marked with 'use client' meant that if the 'use client' directive was missed out then the application would not run without an error being thrown out which would then need to be debugged. The watch function for the password confirmation validation took a little bit of understanding of React Hook Form to figure out that it was asking for cross field value referencing which was fairly self explanatory in this case. It was quite tricky to make sure that the password complexity regex pattern works as intended. This would take a few attempts to get it right. Each attempt would either not filter out some characters classes that are needed or it would filter out a character class that is allowed. It was not easy to work out but after a few attempts finally it worked.

References

https://legacy.reactjs.org/docs
https://youtu.be/cc_xmawJ8Kg?si=0MxIQZGmbexQf-73

https://nextjs.org/docs/app/building-your-application/routing/linking-and-navigating
https://www.freecodecamp.org/news/how-to-create-forms-in-react-using-react-hook-form/